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Source: *Political Psychology*, DECEMBER 2014, Vol. 35, No. 6 (DECEMBER 2014), pp. 795-812

Published by: International Society of Political Psychology

Stable URL: <https://www.jstor.org/stable/43783822>

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## **Societal Threat to Safety, Compensatory Control, and Right-Wing Authoritarianism**

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*We analyzed directly and indirectly the relationships between societal threat to safety, perceived control, and the increase in right-wing authoritarianism (RWA) in two studies. In Study 1 (national sample of the Italian population, N = 1,169), we performed a longitudinal analysis structured into three waves (January 2003, September 2004, and January 2005). A moderated regression analysis showed that RWA increased from 2003 to 2005 as a function of perceived societal threat to safety more among low- than among high RWA scorers. In experimental Study 2 (Italian university students, N = 131) a moderated mediation model showed loss of perceived control to mediate the relation between societal threat to safety and the increase in RWA, but among low authoritarians only. Limitations, implications, and possible developments of this research are discussed.*

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**KEY WORDS:** right-wing authoritarianism, threat, compensatory control, mediation, moderation

Consistent with the idea that perceiving the social world as characterized by randomness and chaos can be a very stressful experience (Janoff-Bulman, 1989), psychologists from many different theoretical and epistemological positions converge in postulating a basic existential motivation to defend from such perception (e.g., Adler, 1946; Heider, 1958; Lefcourt, 1973). This motivation—which tends to vary across cultures (e.g., Ji, Peng, & Nisbett, 2000) and situations (e.g., Burger, 1989)—leads human beings to perceive themselves as being able to control their environment. Research showed that lack of perceived control leads to high arousal, anxiety, and even to see perceptual patterns in incoherent sets of stimuli (Glass et al., 1973; Luck, Pearson, Madden, & Hewett, 1999; Whitson & Galinsky, 2008).

From a social and political psychology point of view, in an interesting series of articles Kay and colleagues (Kay, Whitson, Gaucher, & Galinsky, 2009; Kay, Gaucher, Napier, Callan, & Laurin, 2008; Kay, Shepherd, Blatz, Chua, & Galinsky, 2011) recently developed and empirically validated the Compensatory Control Mechanism (CCM). Relying on this mechanism, people can cope with the existential threat coming from having—chronically or in the here and now—low levels of perceived control over their environment by endorsing external systems that impose structure and order in their social world. According to Kay and colleagues, the main compensatory external systems are believing in a controlling God and in the government: “Given the clear rules, guidelines, norms, and structure formal systems provide, governments and organizational systems, much like religions, hold the potential to help people imbue their worlds with order and control” (Kay et al., 2008, p. 21). Consistent with Rothbaum, Weisz, and Snyder (1982; see also Morling & Evered, 2006; Rudolph, Denning, & Weisz, 1995), resorting to such compensatory systems should be a beneficial strategy to cope with perceived lack of primary control by helping people to perceive a vicarious control over their world. According to them, the most efficient strategy people can rely on—whenever their attempts to directly control their world fail—is to submit to powerful others.

The literature on perceived control shows some intriguing commonalities with the most recent conceptions about right-authoritarianism (RWA), i.e., the covariation of three attitudinal clusters: (1) *authoritarian submission* (a strong tendency to submit to authorities, which are perceived as established and legitimate in the society in which one lives); (2) *authoritarian aggression* (a general aggressiveness, perceived to be positively sanctioned by established authorities, directed against various people); and (3) *conventionalism* (a strong tendency to adhere to the social conventions, which are perceived as endorsed by the society and its established authorities) (Altemeyer, 1996).

Indeed, according to the literature on the CCM, people lacking perceived control over their social world can compensate for such perception by adhering to religion and government, i.e., to established societal authorities (Kay et al., 2008). According to the literature on authoritarianism, RWA accounts for people’s tendency to do so (Altemeyer & Hunsberger, 1993). Moreover, according to Duckitt (2001), RWA should not be considered as a stable personality characteristic; on the contrary, it should be conceived as an ideological variable expressing situational and dispositional motivational goals of order, social control, and security. This dynamic conception of RWA was empirically validated in a five-month longitudinal study performed by Sibley, Wilson, and Duckitt (2007), in which RWA significantly increased as a function of perceiving the world as a threatening place.

According to the literature, situational threat elicits out-group derogation among authoritarians (see the differential moderation hypothesis, Duckitt & Sibley, 2009). However, we argue that such threat can produce an increase in RWA as a compensatory control mechanism among people with low levels of authoritarianism. We based this idea on Dallago and colleagues’ studies (Dallago & Roccato, 2010; Dallago, Mirisola, & Roccato, 2011, 2012), which showed that perceived societal threat stemming from criminality—which undermines people’s perceived control over their social world (Jackson, 2011; Perry & Sibley, 2010)—increased RWA among people with high, but not among those with low, Openness to experience. RWA and Openness to experience are not synonymous; however, they systematically show significant, although low, negative correlations (see Akrami & Ekehammar, 2006; Altemeyer, 1996; Duriez & Soenens, 2006; Ekehammar, Akrami, Gylje, & Zakrisson, 2004; Heaven & Bucci, 2001; Lippa & Arad, 1999; Peterson & Lane, 2001; Peterson, Smirles, & Wentworth, 1997; Van Hiel, Cornelis, & Roets, 2007; Van Hiel, & Mervielde, 2004). Thus, Dallago and colleagues’ studies support the idea that people with low RWA levels (which tend to be high in Openness to experience) may try to compensate the feeling of uncontrollability of the social world stemming from being exposed to threats such as criminality by increasing their level of authoritarianism, in order to be provided with a set of strong and explicit moral norms, prescribing how the society should work.

To summarize, based on the literature above, we assumed that RWA could be a strategy people rely on to compensate for lack of perceived control stemming from societal threat to safety. Moreover, we supposed that resorting to such strategy should be particularly appealing for people low versus high in RWA, as they should have a higher need to compensate threat and lack of personal control. We empirically tested these ideas in two studies. In longitudinal Study 1, we tested the hypothesis that, under conditions of threat to safety, RWA should increase among low RWA scorers, but not among high RWA scorers. In experimental Study 2, we tested an expanded version of this hypothesis, linking threat, perceived control, and RWA.

## Study 1

### *Goals and Hypothesis*

In Study 1 we aimed to answer the following research question: Does RWA actually change, *within the same individuals*, as a consequence of perceived societal threat to safety? Based on the literature above, we hypothesized RWA to increase under conditions of perceived threat to safety among low RWA scorers, but not among high RWA scorers.

### *Method*

We performed a secondary analysis of the longitudinal data collected by the Osservatorio del Nord Ovest (North-Western Observatory), a research institute of the University of Torino, between January 2003 and January 2005. Since 2002, a panel extracted from the Italian population over 14 years old was surveyed three times a year about a number of social issues. In the following three waves, data on RWA and on perceived societal threat to safety were available: (1) January 2003 ( $N = 5,545$ , response rate = 43.0%), (2) September 2004 ( $N = 5,558$ , response rate = 68.5%), and (3) January 2005 ( $N = 4,793$ , response rate = 62.9%).

Each wave's participants have been interviewed via mail and were representative of the Italian population according to gender, age, and education. However, due to the panel attrition which inevitably characterizes longitudinal research (see Ribisl et al., 1996; Tourangeau & Ye, 2009), just 1,169 people participated in all of these three waves (women: 42.08%, mean age = 52.38,  $SD = 15.68$ , mean years of education = 10.89,  $SD = 3.81$ ). The present study was thus focused on them. As concerns age and education, this sample nicely approximated the Italian population, compared to which our participants were just 2.94 years older and showed 1.74 more years of education, even if women were somewhat underrepresented (42.08 in our sample vs. 52.15 in the Italian population).

### *Measures*

In all the waves, participants' gender, age, and education were available. To keep under control the heterogeneity of our dataset, much higher than that characterizing the mainstream research about RWA, which is typically performed on student samples, we used them as control variables, taking into account their value in January 2003.

In January 2003, four five-category items from Giampaglia and Roccato's (2002) Italian version of Altemeyer's (1996) RWA scale were available. Given their small number, it is far from surprising that their alpha was (slightly) below the usual .70 threshold,  $\alpha = .63$ . However, their mean correlation, *mean*  $r = .30$ , was satisfactory and higher than those stemmed from previous Italian research (see, for example, Dallago, Cima, Roccato, Ricolfi, & Mirisola, 2008; Dallago & Roccato, 2010;

Dallago et al., 2011, 2012; Giampaglia & Roccato, 2002; Mirisola, Sibley, Boca, & Duckitt, 2007). Thus, we computed a synthetic RWA index at T<sub>1</sub>, as the mean of these four items, ranging from 1.00 to 5.00.

In January 2005, six more five-category items from the same scale were available,  $\alpha = .72$ . Again, we computed a synthetic RWA index at T<sub>2</sub>, as the mean of these six items, ranging from 1.00 to 5.00. The items we used to assess RWA in our second wave were different from those we used in our first wave. However, a pilot study, performed on 353 students at the University of Torino, previously showed that the correlation of our operationalizations of RWA at T<sub>1</sub> and of RWA at T<sub>2</sub> was a satisfactory  $r = .80, p < .001$ . Thus, we considered them as parallel forms of the same measurement tool.

As concerns perceived societal threat to safety, we took from the September 2004 wave the item “Think of micro-criminality: How would you define the situation regarding this problem in Italy?” This four-category item (1 = not dangerous at all; 4 = very dangerous) was Dallago and Roccato’s (2010) operationalization of perceived societal threat to safety.

Data Analysis

To test whether RWA at T<sub>1</sub> moderated the effect exerted on RWA at T<sub>2</sub> by perceived threat, we conducted a hierarchic moderated multiple regression (Aiken & West, 1991; Cohen, Cohen, West, & Aiken, 2003) using the DeducerMMR R plugin (Mirisola & Seta, 2011; Mirisola, Seta, Gentile, & La Guardia, 2011). In the first step, we entered our control variables, RWA at T<sub>1</sub>, and perceived societal threat to safety. We dummy-coded gender (1 = woman), while we centered the other predictors. In the second step, we added the interaction between RWA at T<sub>1</sub> and perceived societal threat.

Results

Table 1 shows the descriptive statistics for our variables and their correlations. As shown in Table 2, in the first step of our regression RWA at T<sub>1</sub>, age, education, and perceived societal threat to safety significantly fostered RWA at T<sub>2</sub>,  $R^2 = .36$ . After adding the first-order interaction between RWA at T<sub>1</sub> and perceived threat, the model’s  $R^2$  rose to .37. This increase was statistically significant,  $F(1, 1162) = 7.05, p < .01$ . Based on Cohen et al. (2003), to depict this interaction graphically we first z-scored the RWA change. Subsequently, we estimated the means for participants low (−1 SD) and high (+1 SD) in RWA at T<sub>1</sub> (see Figure 1). Simple slopes analysis showed that perceived societal threat to safety strongly predicted RWA at T<sub>2</sub> among participants with low levels of authoritarianism at T<sub>1</sub>, simple slope = .278,  $t(1162) = 6.41, p < .001$ , and much more weakly at high levels of

Table 1. Study 1: Descriptive Statistics for Our Variables and Correlations Among Them

|                                                         | <i>M</i> | <i>SD</i> | 1    | 2       | 3       | 4       | 5       | 6       |
|---------------------------------------------------------|----------|-----------|------|---------|---------|---------|---------|---------|
| 1. Woman                                                | 42.08    | .49       | 1.00 | −.11*** | −.04    | −.04    | .01     | .08*    |
| 2. Age                                                  | 52.38    | 15.68     |      | 1.00    | −.19*** | .18***  | .17**   | .08**   |
| 3. Years of education                                   | 10.89    | 3.81      |      |         | 1.00    | −.15*** | −.17*** | −.10*** |
| 4. RWA T <sub>1</sub> (January 2003)                    | 3.35     | .86       |      |         |         | 1.00    | .58***  | .23***  |
| 5. RWA T <sub>2</sub> (January 2005)                    | 3.02     | .82       |      |         |         |         | 1.00    | .29***  |
| 6. Perceived societal threat to safety (September 2004) | 3.34     | .59       |      |         |         |         |         | 1.00    |

\* $p < .05$ , \*\* $p < .01$ , \*\*\* $p < .001$ .

Table 2. Study 1: Predictors of RWA at T<sub>2</sub>

|                                                             | Model 1  |             |      | Model 2  |             |      |
|-------------------------------------------------------------|----------|-------------|------|----------|-------------|------|
|                                                             | <i>B</i> | <i>S.E.</i> | Beta | <i>B</i> | <i>S.E.</i> | Beta |
| Intercept                                                   | 3.00***  | .02         |      | 3.01***  | .03         |      |
| Woman                                                       | .04      | .04         | .02  | .04      | .04         | .03  |
| Age                                                         | .00*     | .00         | .05  | .00*     | .00         | .06  |
| Education                                                   | −.01**   | .01         | −.06 | −.01*    | .00         | −.06 |
| RWA at T <sub>1</sub> (January 2003)                        | .51***   | .02         | .53  | .50***   | .02         | .52  |
| Perceived societal threat to safety (September 2004)        | .21***   | .03         | .15  | .20***   | .03         | .15  |
| RWA at T <sub>1</sub> * Perceived societal threat to safety |          |             |      | −.09**   | .03         | −.07 |

\**p* < .05, \*\**p* < .01, \*\*\**p* < .001.

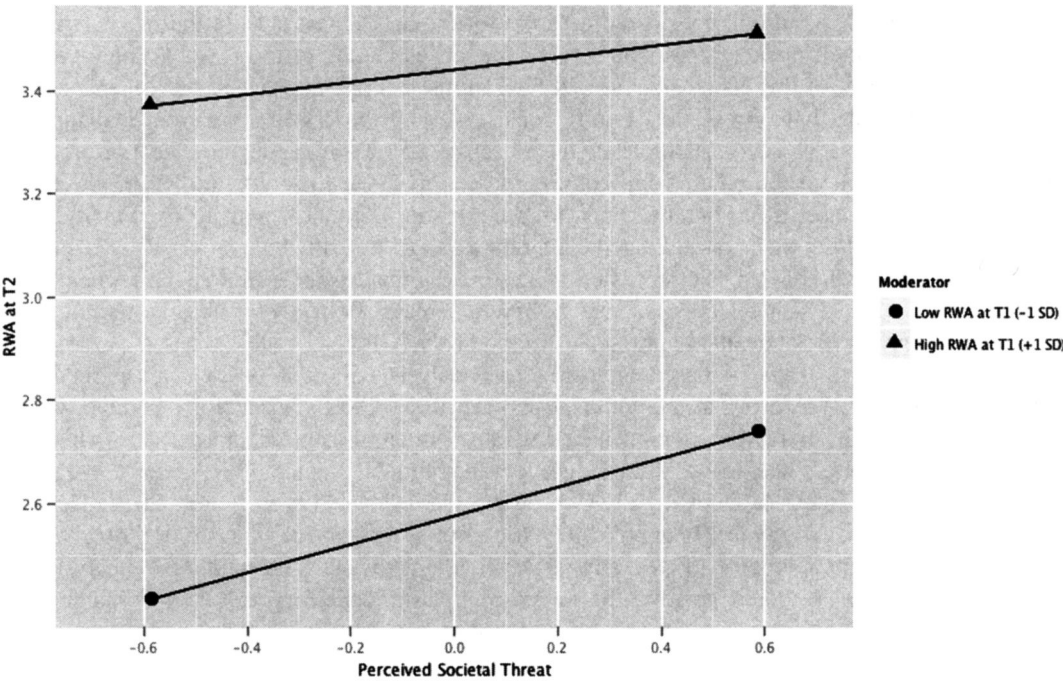


Figure 1. Study 1: Moderating effect of RWA at T<sub>1</sub> on the relation between perceived threat to safety and RWA at T<sub>2</sub>.

authoritarianism at T<sub>1</sub>, simple slope = .118, *t*(1162) = 2.51, *p* < .05.<sup>1</sup> The moderation effect we detected was plausibly not the consequence of a ceiling effect displayed among high RWA scorers. Indeed, just the 0.8% of our participants showed the theoretical maximum of RWA at T<sub>1</sub>. Most importantly, parallel analyses performed using the Tobit model (Austin, Escobar, & Kopec, 2000; McBee, 2010)—which leads to robust results when managing floor or ceiling effects<sup>2</sup>—gave results (available upon request) identical to those we presented.

<sup>1</sup> Based on Mavor, Louis, and Sibley (2010), we performed a confirmatory factor analysis in which RWA was modeled with three-item parcels (conventionalism, authoritarian aggression, and authoritarian submission, respectively). The model showed a good fit (*CFI* = .97, *TLI* = .93, *RMSEA* = .07). Thus, we conducted parallel analyses measuring RWA by giving equal weight to the items assessing the construct’s three attitudinal clusters. Obtained results, available upon request, were substantially identical to those we presented.

<sup>2</sup> We based this on Cox and Oakes (1984) and chose the RWA empirical maximum as censoring point.



## *Discussion*

In this study, we tested the hypothesis that in the condition of perceived societal threat to safety, low—but not high—RWA scorers increase their RWA level as a consequence of societal threat to safety. Results revealed that both low and high RWA scorers increased their RWA level in conditions of perceived threat. However, among the latter, the increase was much weaker than among the former and plausibly reached statistical significance just because of the large *N* of our dataset. Thus, as a whole, we conclude that our hypothesis was substantially verified.

These results were fully consistent with Van Hiel and De Clercq (2009), according to which RWA should be considered as a resource people may efficiently use to cope with stress. Indeed, these authors found that RWA reduced both the impact of a distressed personality on depression and on the negative consequences of 21 potentially stressful life events that participants actually experienced in the 24 months preceding the survey. Discussing these findings, Van Hiel and De Clercq (2009) concluded that RWA should be considered a “good for the self” construct,<sup>3</sup> in that it “buffers the effects of facilitators of mental distress on actual experienced mental distress” (p. 47; however, for results inconsistent with this conclusion, see Duriez, Klimstra, Luyckx, Beyers, & Soenens, 2012). Moreover, our results have been substantially consistent with those stemming from Dallago and colleagues’ (Dallago & Roccato, 2010; Dallago et al., 2011, 2012) converging cross-section analyses on the moderator role of the relationship between perceived threat to safety and RWA, played by Openness to experience, a personality trait which negatively correlates with RWA. Finally, they indirectly supported the idea of linking the literature on RWA with that on the CCM, in that psychological research showed that threat stemming from criminality undermines people’s perceived control over the social world (Jackson, 2011; Perry & Sibley, 2010). Unfortunately, measures of perceived control were not available in the dataset we analyzed.<sup>4</sup>

In conclusion, this study—based on secondary analysis—allowed us to analyze the links between threat and RWA using a longitudinal approach in a wide Italian national sample, which closely resembled the Italian population in terms of sociodemographic characteristics. This was particularly intriguing, since, to our knowledge, few longitudinal studies on RWA have been performed (Asbrock, Sibley, & Duckitt, 2010; Asbrock, Christ, Duckitt, & Sibley, 2012; Heaven, Ciarrochi, & Leeson, 2011; Liu, Huang, & McFedries, 2008; Peterson & Lane, 2001; Sibley et al., 2007), and none of them empirically test any moderation hypothesis. Our longitudinal analysis has had a two-year span. It would probably be more appropriate to use shorter intervals among the waves. However, even with such long intervals, we have found theoretically sound outcomes: In our opinion, this affirms the robustness of our results. Moreover, the field nature of our data and the high heterogeneity of our sample gave our results pretty strong ecological validity. However, on the negative side, we could not measure participant’s perceived control over their social world, and thus we gained only indirect evidence on the plausibility of linking the literature on RWA to

<sup>3</sup> For a conception of RWA as a “good for the in-group” construct, see Kessler and Cohrs (2008).

<sup>4</sup> It should be noted that in this study we worked on perceived, and not on actual, threat to safety. This should not be considered a strong limitation, as ecological analysis shows that variables expressing concern about crime as a social problem are strongly correlated to the actual crime spread in the area people live in (e.g., Miceli, Roccato, & Rosato, 2004). As a matter of fact, in parallel analyses (results available upon request), we tried to employ the high quality of our dataset testing a multilevel model aimed at analyzing the effect exerted on the increase in RWA by the cross-level interaction between RWA at *T*<sub>1</sub> and the crime rate of the area where our participants lived in. Unfortunately, such a cross-level interaction did not gain statistical significance. However, the structure of our data allowed us to use level-2 data at a plausibly suboptimal level, i.e., at the county level (in Italy there are 110 counties, which are local governmental entities whose jurisdictional territories usually include numerous towns geographically surrounding one main city). We believe that this level of aggregation—which mixed areas characterized by very different actual crime rates (Russo, Roccato, & Vieno, 2011)—was far from satisfactory. In fact, multilevel analysis tends to be used at the block or at the neighborhood level (e.g., Perkins, Meeks, & Taylor, 1992) in order to take into account homogeneous ecological clusters. Thus, before closing the book on the multilevel prediction of RWA, more appropriate ecological data should be used.

that of the CCM. To overcome these limitations, we performed Study 2 using an experimental approach.

## Study 2

### *Goals and Hypothesis*

This study tackled the following research question: Does loss of perceived control over participants' social world help explain and qualify the effect found in Study 1? Based on the literature above, we hypothesized that (1) threat should foster loss of perceived control, and (2) loss of control perception should lead to an RWA increase among participants with initial low levels of RWA. Thus, we hypothesized that participants with initial low, but not those with initial high, RWA scores, will bolster societal order (Shepherd, Kay, Landau, & Keefer, 2011) by increasing their RWA level as a function of the decrease of perceived control stemming from threat.

### *Method*

We built an experiment using Lau and Redlawsk's (2001) Dynamic Process Tracking Environment (DPTE). The DPTE is a computer-based dynamic-information board developed to study decision making in complex social situations, originally developed and applied to the study of electoral campaign environments. The DPTE is a revision of the traditional static-information board developed to gain more external validity by better mimicking the flow of information in a real-world social context. This technique allows for tracing the decision-making process as it happens while the information label scrolls down on a computer screen: Thus, a limited number of labels are visible at any time. Participants can access the information they are interested in by clicking on the label. While reading the detailed information, the scrolling continues in the background.

We chose to collect our data by the DPTE to disclose our goals and to provide our study with external validity. We used the DPTE to simulate a mock electoral campaign. Our experimental procedure included four main stages: (1) a preexperimental questionnaire; (2) a two-minute practice session; (3) a nine-minute mock electoral campaign; and (4) a postexperimental questionnaire. The electoral campaign (see details below) we simulated was built based on Redlawsk (2004; Redlawsk, Civettini, & Emmerson, 2010).

### *Participants*

From the University of Torino, Italy, 131 students (49 men, mean age = 22.76,  $SD = 5.51$ ) participated in our study. Of these, 90 completed the tasks in a social psychology lab, while the other 41 did so while at home by connecting to the website <http://dpte.polisci.uiowa.edu/>, where the DPTE can be used. Preliminary analyses showed that this design variable did not affect our results. Thus, we did not take it into consideration in the analyses presented.

### *Preexperimental Questionnaire<sup>5</sup>*

RWA at  $T_1$  was assessed using 10 four-category balanced items randomly chosen from those of Giampaglia and Roccato's (2002) Italian version of Altemeyer's (1996) RWA Scale. Based on  $\alpha = .72$ , we computed participants' RWA score at  $T_1$  as the mean of these items, which ranged between 1.10 and 3.00.

<sup>5</sup> Beyond those we presented in this section, in the pre- and the postexperimental questionnaires we asked our participants more questions. Readers interested in examining the full questionnaires may ask for them from the corresponding author.



Perceived control at  $T_1$  was assessed by the six items used by Kay et al. (2008; e.g., “To a great extent my life is controlled by accidental happenings” and “The things that occur in my life are mostly a matter of chance”). Based on  $\alpha = .78$ , we computed participants’ perceived control in their life as the mean of these six items. High scores expressed high perceived control.

We used in the supplementary analyses Dangerous World Beliefs at  $T_1$ , (see below), which were assessed by averaging 10 balanced items ( $\alpha = .81$ ) randomly chosen from the Italian version of Altemeyer’s Dangerous World Beliefs Scale (Mirisola, Di Stefano, & Falgares, 2007).

### *Experimental Session*

We created a mock electoral campaign with four candidates competing for the role of Italian Prime minister in the 2020 general election. After reading an initial announcement (see Appendix A), participants experienced the political campaign; information about the candidates running for the election as well as nonpolitical information (such as generic information about the country) was available. We introduced the experimental manipulation in the middle of the campaign. A randomly selected group of participants ( $n = 62$ ) read a secure scenario, in which Italy in 2020 was presented as one of the most secure nations in the world and the Italians as believing they live in one of the best periods of the human history. The other participants ( $n = 69$ ) read an insecure scenario, which depicted Italy in 2020 as a very dangerous place, in which home burglaries are a common experience, and people avoid walking alone at night because armed squads control many city districts and regularly engage in assault and robbery. The scenarios’ full texts are presented in Appendix B. Right after the experimental manipulation, the electoral campaign continued. At the end of the campaign, participants were asked to cast their vote.

### *Postexperimental Questionnaire*

RWA at  $T_2$  was assessed using other 10 balanced items randomly chosen from Giampaglia and Roccato’s (2002) Italian version of Altemeyer’s (1996) RWA Scale. Based on  $\alpha = .79$ , we computed participants’ RWA at  $T_2$  scores as the mean of such 10 items. The scores ranged from 1.00 to 2.80. A pretest, performed on 242 students from the University of Torino, showed that our two RWA measures could be considered as parallel forms of the same measurement tool,  $r = .81$ ,  $p < .001$ .

Participants’ perceived control of their life at  $T_2$  was measured using the same six items we used in the preexperimental session. Based on  $\alpha = .89$ , we computed participants’ scores as the mean of these items. Building on Williams, Zimmerman, Rich, and Steed (1984a, 1984b), in our analyses we examined the mediating effect of the loss of perceived control using perceived control’s residual-gain score estimate. The residual-gain score estimate is the difference between an actual and a predicted dependent variable (control at  $T_2$ ), computed as the residual of a linear regression equation with the pretest score (in our case, perceived control at  $T_1$ ) as predictor variable and the posttest score (in our case, perceived control at  $T_2$ ) as dependent variable.

As a manipulation check, we asked our participants to answer the item on perceived societal threat to safety we used in Study 1. Moreover, we asked them to answer 10 other balanced items randomly chosen from Mirisola and colleagues (2007) Italian version of Altemeyer’s Dangerous World Beliefs Scale. Based on  $\alpha = .92$ , we averaged them to compute our participants’ Dangerous World Beliefs scores. This scale has also been used to calculate residual-gain scores of Dangerous World Beliefs between  $T_1$  and  $T_2$  (see below).

Finally, we asked our participants to fill in a standard sociodemographic form.

Table 3. Study 2: Descriptive Statistics for Our Variables and Correlations Among Them

|                          | <i>M</i> | <i>SD</i> | 1 | 2   | 3     | 4     |
|--------------------------|----------|-----------|---|-----|-------|-------|
| 1. Scenario              |          |           | 1 | .03 | .08   | -.20* |
| 2. RWA at T <sub>1</sub> | 1.79     | .38       |   | 1   | .80** | -.07  |
| 3. RWA at T <sub>2</sub> | 1.72     | .40       |   |     | 1     | -.09  |
| 4. Control change        | .00      | .49       |   |     |       | 1     |

\**p* < .05, \*\**p* < .001.

Table 4. Study 2: Regression Results for Conditional Indirect Effect

|                                                   | <i>B</i> | <i>SE</i> | Beta |
|---------------------------------------------------|----------|-----------|------|
| <i>Control-perception change</i>                  |          |           |      |
| Constant                                          |          | .04       | .00  |
| Secure vs. insecure scenario                      | -.20*    | .04       | -.10 |
| <i>RWA at T<sub>2</sub></i>                       |          |           |      |
| Constant                                          |          | .02       | -.01 |
| Secure vs. insecure scenario                      | .04      | .02       | .02  |
| Control-perception change                         | -.07     | .05       | -.06 |
| RWA at T <sub>1</sub>                             | .77**    | .06       | .81  |
| RWA at T <sub>1</sub> * Control-perception change | .13*     | .14       | .33  |

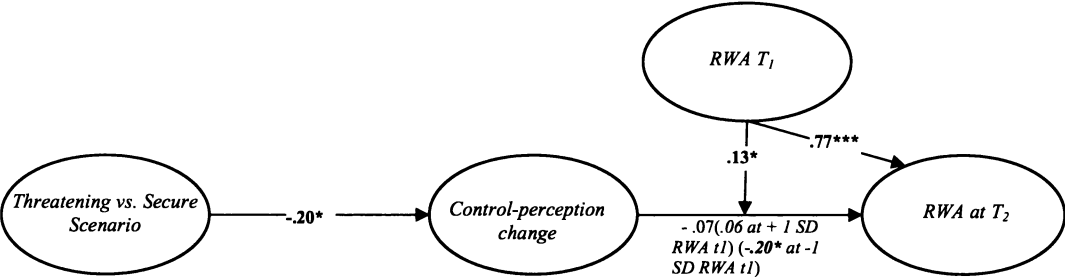
\**p* < .05, \*\**p* < .001.

Results

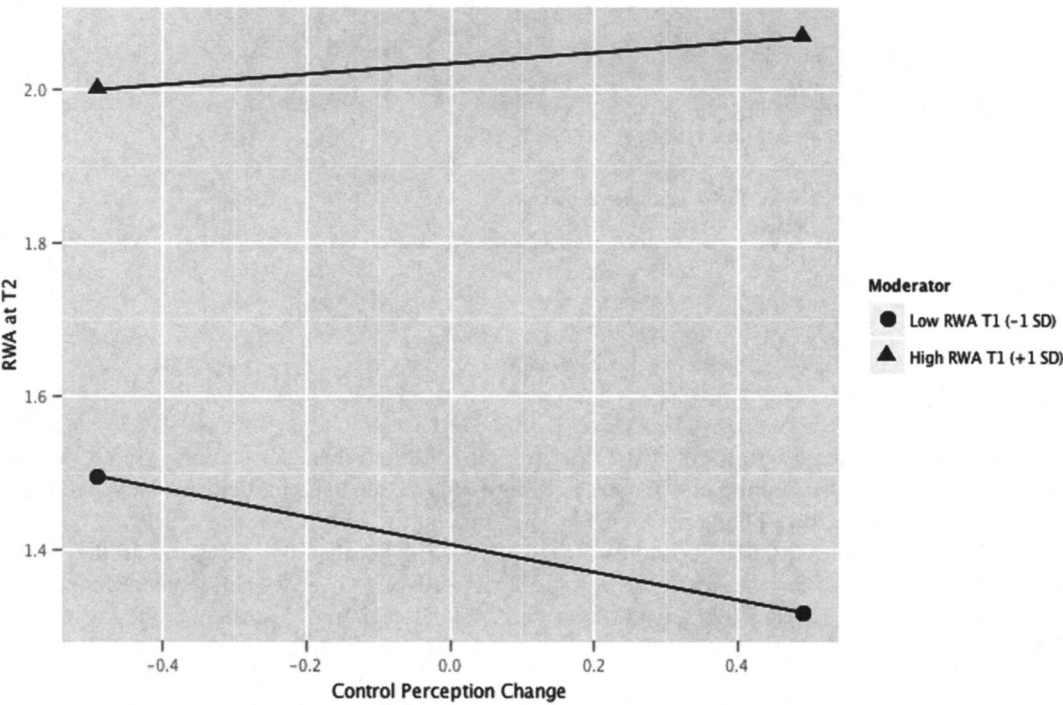
Table 3 shows the descriptive statistics for our variables and their correlations. Before testing our hypotheses, we checked the effectiveness of our manipulation. The participants exposed to the threatening scenario showed higher perceived societal threat to safety scores (*M* = 3.19, *SD* = 2.05) and higher Dangerous World Belief scores (*M* = 4.43, *SD* = 1.12) than those exposed to the secure scenario (*M*'s = 2.05, *SD* = .50, and 2.57, *SD* = .53, respectively). The two differences were statistically significant, *t*(129) = 6.096, *p* < .001,  $\eta^2$  = .224, and *t*(129) = -5.772, *p* < .001,  $\eta^2$  = .205. Thus, we concluded that our manipulation actually influenced participants' sense of menace in the expected direction.

Using Hayes' (2012) PROCESS macro (model 14), a moderated mediation analysis (Preacher, Rucker, & Hayes 2007) was performed in order to assess the indirect effect of being exposed to the insecure versus the secure scenario on the RWA increase through the mediation of the change of perceived control over the social world, at different RWA levels at T<sub>1</sub> (i.e., at  $\pm 1$  standard deviation from the mean). Based on Cohen et al. (2003), we centered our continuous variables, while the variable expressing being exposed to the threatening versus the secure scenario was included using an unweighted effect coding. Regression results for our moderated mediation model are reported in Table 4.

Consistent with our hypothesis (see Figure 2), being exposed to the threatening versus the secure scenario predicted a reduction of perceived control, which, in turn, significantly influenced RWA at T<sub>2</sub>. However, the latter effect was moderated by RWA at T<sub>1</sub>. Simple slopes analysis showed that the loss of perceived control fostered RWA at T<sub>2</sub> among low RWA scorers at T<sub>1</sub>, simple slope = -0.18, *t*(126) = -2.25, *p* < .05, but not among participants who, at T<sub>1</sub>, scored high in RWA, simple slope = 0.07, *t*(126) = 1.21, *p* = .23 (see Figure 3). Bootstrapping estimation (5,000 bootstrap



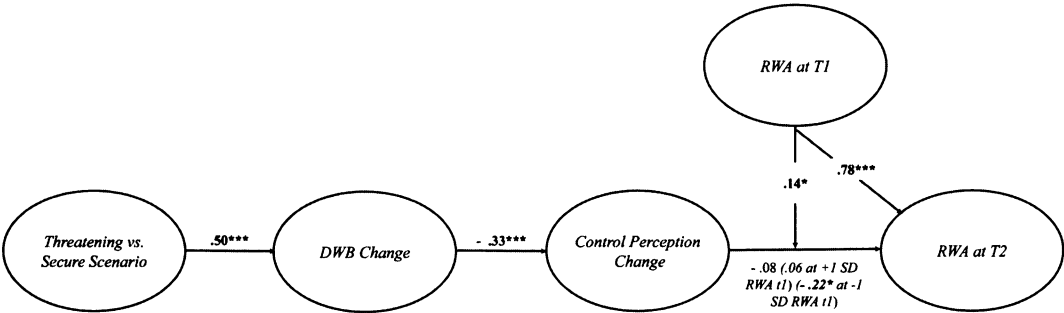
**Figure 2.** Study 2: Moderated mediation model predicting RWA at T<sub>2</sub> as a function of manipulated threat, control-perception change, and RWA at T<sub>1</sub> (standardized betas are displayed).



**Figure 3.** Study 2: Moderating effect of RWA at T<sub>1</sub> on the relation between control-perception change and RWA at T<sub>2</sub>.

samples) showed a significant indirect effect exerted by threat for participants with low RWA at T<sub>1</sub> (95% CI .003, .047), whereas no indirect effect was found for participants with high RWA at T<sub>1</sub> (95% CI -.030, .004).<sup>6</sup>

<sup>6</sup> Parallel analysis (performed using model 59 of the PROCESS macro) showed that RWA at T<sub>1</sub> did not moderate both the relations between being exposed to the threatening-versus-secure scenario and RWA increase and between being exposed to the threatening-versus-secure scenario and loss of perceived control, respectively (results available upon request). Moreover, like in Study 1, a confirmatory factor analysis of a three-dimensional model based on Mavor et al. (2010) model showed a good fit (*CFI* = .96, *TLI* = .92, *RMSEA* = .10). Subsequent parallel analyses, performed giving equal weight to the items assessing RWA's three attitudinal clusters, led to results, available upon request, substantially analogous to those we chose to present.



**Figure 4.** Study 2: Supplementary analyses—Moderated mediation model predicting RWA at T<sub>2</sub> as a function of manipulated threat, dangerous word beliefs change, control-perception change, and RWA at T<sub>1</sub> (standardized betas are displayed).

As in Study 1, the moderation effect we have detected did not seem to be the consequence of a ceiling effect displayed among high RWA scorers. Indeed, top RWA scorers showed an empirical maximum of RWA (3.00 and 2.80 at T<sub>1</sub> and T<sub>2</sub>, respectively) lower than the theoretical maximum of the scale (4), with a maximum increment of .70 between RWA at T<sub>1</sub> and RWA at T<sub>2</sub>. Most importantly, parallel analyses, performed using the Tobit model (Austin et al., 2000; McBee, 2010)<sup>7</sup> led to results (available upon request) analogous to those presented.

*Supplementary Analyses*

In the supplementary analyses—conducted on the request of an anonymous reviewer—we could not replicate directly the effect we detected in Study 1. Indeed, RWA at T<sub>1</sub> did not moderate the relationship between scenario and RWA at T<sub>2</sub> when we used the threatening-versus-secure scenario as a predictor,  $p = .79$ . This result was far from surprising, given that previous research showed that threat, when experimentally manipulated via a scenario depicting future conditions of the nation where participants lived, influenced RWA not directly, but via the mediation of dangerous world beliefs (Duckitt & Fisher, 2003).

To test this interpretation, we explicitly integrated Duckitt’s (2001) and Kay and colleagues’ (2008) models. To do so, we first replicated the interaction effect of Study 1 using the residual gain score of DWB as predictor on the 130 participants who answered the DBW scale both in the preexperimental and in the postexperimental questionnaires. In the first step of our moderated regression, RWA at T<sub>2</sub> was significantly predicted by RWA at T<sub>1</sub>,  $\beta = .80$ ,  $t(126) = 14.81$ ,  $p < .001$ , but not by the DWB change,  $\beta = .04$ ,  $t(126) = 0.77$ ,  $p = .44$ . After adding the first-order interaction between RWA at T<sub>1</sub> and DWB change, which significantly influenced RWA at T<sub>2</sub>,  $\beta = -.11$ ,  $t(125) = -1.99$ ,  $p < .05$ , the model’s  $R^2$  rose to .65. This increase was statistically significant,  $F(1, 125) = 3.95$ ,  $p < .05$ . As in Study 1, simple slopes analysis showed that DWB predicted RWA at T<sub>2</sub> among participants with low levels of authoritarianism at T<sub>1</sub>, simple slope = .15,  $t(125) = 2.06$ ,  $p < .05$ , but not at high levels of authoritarianism at T<sub>1</sub>, simple slope = -.04,  $t(125) = -.73$ ,  $p = .47$ .

Then, using MPlus (ML estimator), we reanalyzed the model depicted in Figure 2 entering the DWB residual-gain score as mediator of the relation between threatening scenario and control-perception change. The new model (see Figure 4) showed a good fit,  $CFI = .96$ ,  $TLI = .93$ ,  $RMSEA = .10$ . The bootstrapping (5,000) estimate of the indirect effect showed that the DWB change fully mediated the relationship between threatening scenario and control-perception change, indirect effect = -.16, 95%  $CI$ : -.27, -.06. Moreover, as happened in the model depicted in Figure 2, RWA

<sup>7</sup> Based on Cox and Oakes (1984), we chose the RWA empirical maximum (2.8) as the censoring point.

at  $T_1$  moderated the relationship between control-perception change and RWA at  $T_2$ . No significant interaction effects on RWA at  $T_2$  were found between RWA at  $T_1$  on the one hand and DWB change and threatening-versus-secure scenario on the other hand (results available upon request).

### *Discussion*

In this study, we have fine-tuned the results from Study 1 and those available in the literature (Dallago et al., 2011, 2012; Dallago & Roccato, 2010; Shepherd et al., 2011), showing that the link between societal threat and RWA at  $T_2$ , beyond being moderated by people's initial level of RWA, is mediated by a reduction in perceived control. As a whole, consistent with our hypothesis, experimentally induced societal threat reduced participants' perceived control, which in turn fostered RWA among low, but not among high, RWA scorers. These results have been the first ones linking explicitly the literature on the origins of RWA with that on the Compensatory Control Mechanism (Kay et al., 2008), helping us to understand the processes in action when people try to cope with the distress stemming from living in a threatening world by increasing their RWA level. Due to the fact that our manipulation threat was embedded in an environment that was much more complex than those usually used in experimental research on RWA (e.g., Altemeyer, 1988; Duckitt & Fisher, 2003; Sales & Friend, 1973), we feel confident about the ecological validity of obtained results.

### **General Discussion**

In this research, we aimed at linking theoretically and empirically two literatures: First is the Compensatory Control Model (Kay et al., 2008; Kay et al., 2009; Kay et al., 2011), according to which people's motivation to perceive themselves as able to control their social world leads them to resort to compensatory external sources of control (mainly God and the government) when feeling unable to control their world directly. Second, is the most recent literature on the links between threat and right-wing authoritarianism, according to which the latter construct should be considered as a "good for the self" (Van Hiel & De Clercq, 2009) ideological variable (Duckitt, 2001), which may change its level as a function of (perceived or actual) societal threat (Sibley et al., 2007) mainly among low RWA scorers (Dallago & Roccato, 2010; Dallago et al., 2011, 2012). In Study 1, we showed perceived societal threat to safety to lead to an RWA increase mainly among low RWA scorers. In Study 2, we showed the change in perceived control over participants' social world to mediate the link between threat and RWA among low RWA scorers only.

In spite of relevant methodological differences (Study 1 was based on a two-year longitudinal field survey, while Study 2 was based on an experiment performed in a psychology lab) and different operationalization of the threat we used (in Study 1, we measured perceived societal threat to safety, while in Study 2, we experimentally manipulated such threat), our studies led to converging results. This supported the robustness of the conclusions we have drawn. Moreover, our methodological approach utilizing the same participants allowed us to assess the changes in RWA and the changes in perceived control (Study 2). As a whole, our study showed that RWA can be considered as a card people (especially those initially low in this variable) can strategically choose to play in response to stressful and/or threatening external circumstances that decrease control perception.

Studies performed on the differential moderation hypothesis (Duckitt & Sibley, 2009) showed situational threat fosters out-group derogation among high, but not among low, RWA scorers (e.g., Cohrs & Asbrock, 2009; Cohrs & Ibler, 2009; Dru, 2007; Feldman & Stenner, 1997; Rickert, 1998; Stenner, 2005). We believe our result, far from contradicting them, should be considered as complementary to those from previous research. Indeed, our studies and those testing the differential moderation hypothesis analyzed different outcomes, i.e., RWA at  $T_2$  and prejudice, respectively. Moreover, in Study 2, we showed threat fosters a decrease in perceived control among both high and



low RWA scorers. The literature shows people high in RWA trying to cope with situational menace resorting to a heightened intergroup bias, which can be considered the consequence of “self-protective, defensive motivational needs for control and security” (Duckitt, 2001, p. 85). Our studies showed that people low in RWA try to compensate for the loss of control stemming from threat by increasing their RWA level, i.e., becoming similar to their authoritarian counterparts. Future longitudinal research, performed by repeatedly threatening high and low RWA scorers and measuring both their RWA trend and their prejudice towards out-groups, will be germane.

Four main conclusions can be drawn from our research. First, as concerns RWA, obtained results were consistent with the conception (see Duckitt, 2001) of this construct as an ideological variable and inconsistent with the conception of authoritarianism as a stable personality (or character) trait, which, beyond characterizing the first studies on this topic (Adorno, Frenkel-Brunswick, Levinson, & Sanford, 1950; Fromm, 1941; Reich, 1933), at present is still widely found in the literature (e.g., Altemeyer, 1996). Second, as concerns the origins of RWA, our research, focusing on perceived control as mediator and the initial RWA level as a moderator of the links between threat and RWA, helped fine-tune the literature on the threat-RWA relation (e.g., Altemeyer, 1988; Feldman & Stenner, 1997; Rickert, 1998; Sales & Friend, 1973; Stevens, Bishin, & Barr, 2006). Third, given that RWA accounts for people’s tendency to submit to societal authorities (Altemeyer, 1996), our study expands the literature on the CCM by suggesting that God and the government—i.e., the two sources of external control which, according to Kay and colleagues (2008), people may resort to in order to gain compensatory control over their social world—should be considered as special cases of a more general, external source of perceived control, represented by established authorities. Fourth, we have confirmed the idea that moderation analyses may improve research on authoritarianism: Indeed, according to the methodological literature, the identification of moderators of the relationship between variables indicates the degree of sophistication and maturity of a field of investigation (Aguinis, Boik, & Pierce, 2001; Judd, McClelland, & Culhane, 1995). Some years ago, Sibley and Duckitt (2008) underscored the need for research on the interactions between the predictors of RWA. This article could be considered as a step in satisfying this need.

The conception of RWA as a psycho-political mechanism people can resort to in order to compensate for a severe loss of personal control over their social world resembles Fromm’s (1941) classic idea of authoritarianism as a mechanism people can use to escape from a freedom they feel unable to manage because of the lack of internal resources. However, Fromm conceived authoritarianism as a stable character trait people developed in their childhood as a consequence of a mix of repressive education and of living in a threatening world. On the contrary, our data showed that RWA could vary, within the same individuals, as a consequence of the loss of perceived control due to societal threat (actual and perceived) to safety. In other words, they confirmed Dallago and colleagues’ idea that, under specific societal conditions, authoritarianism differences between low and high RWA scorers tend to disappear. Since RWA accounts for relevant proportions of the variance of generalized prejudice (McFarland, 2011), support of the death penalty, punitive attitudes towards unconventional persons, of approval of the injustice perpetrated by governing authorities, and even of obedience in Milgram-style experiments (Altemeyer, 1981, 1988, 1996), the conception of RWA as a “good for the self” construct (Van Hiel & De Clercq, 2009) sounds rather worrying to us and makes us even more pessimistic than Fromm was.

As often happens, our studies led us to answer some research questions and raised some other questions. First, we could not quantify the length of the raise in RWA we detected in threatened low RWA scorers. Second, we do not know what would happen to our model when tested in field research using a measure of actual criminal threat instead of a variable assessing perceived societal threat to safety. In future research, these two research questions could be answered using multilevel longitudinal-moderated mediation models structured in more than two waves and tested on files aggregated at the block or the neighborhood level.

Moreover, in our two studies, we took into account perceived and actual threat stemming from criminality. What would happen if we were to take into account other forms of threat? In Stenner's (2005) words, threat stemming from criminality should be considered a normative threat, as it menaces the normative order, i.e., to "some system of oneness and sameness that makes 'us' an 'us': some demarcation of people, authorities, institutions, values, and norms that for some folks at some points define who 'we' are, and what 'we' believe in. 'Normative threats' are threats to this oneness and sameness" (p. 17). What would happen if we were to analyze other kinds of normative threats? Working with an Italian sample, Dallago et al. (2011) recently showed that Openness (which negatively correlates with participants' initial RWA level) does *not* moderate the relationship between threat coming from terrorism and RWA, plausibly because RWA is activated as a compensatory mechanism only in the case of threats that are proximal and/or that have been directly experienced, which is not the case of terrorism for a large majority of the Italian population. A development of the present research may focus on other forms of normative and nonnormative threat in order to compare their role in predicting RWA changes.

Finally, among researchers there is no agreement on the "goodness for the self" of RWA: According to Van Hiel and De Clercq (2009) and to Dallago and colleagues (Dallago et al., 2011, 2012; Dallago & Roccato, 2010), RWA is a psychological variable helping people cope with stress, while according Duriez and colleagues (2012), it is not such a psychological variable. As a matter of fact, these researchers' did not assess coping directly: Thus, their conclusions have been merely inductive. Future experimental research performed by analyzing simultaneously and explicitly coping strategies used by low and high RWA scorers facing a threat would be interesting.

To conclude, besides stimulating new questions to be addressed in future research, we believe the present study sheds light on psychological processes underling RWA changes, leading to a better understanding of this complex construct.

## ACKNOWLEDGMENTS

We would like to thank the two anonymous reviewers who evaluated the first version of this article for their insightful suggestions. Correspondence concerning this article should be sent to Michele Roccato, Department of Psychology, University of Torino, Via Verdi 10, 10124 Torino, Italy. E-mail: michele.roccato@unito.it

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## Appendix A

It is 2020 and after some years living abroad you come back to Italy. You fulfilled your life plans, especially as concerns family, job, career and friends. When you come back you find out that Italy has changed a lot. Politically speaking, the same old parties you used to know still exist, even though their leaders changed. In the meantime, many new parties appeared on the political arena. The general election is oncoming and you'll have to cast a vote: The electoral campaign is about to begin. Your task is to get a sense of what your country looks like in 2020 and of the candidates running for the election. Once the campaign is over, you will have the chance to cast a vote.

During an electoral campaign, people get their news from different sources (newspapers, television, internet, friends and relatives, associations, the candidate themselves, and so on). There is much more available than anyone can possibly pay attention to and this will be true in our campaign as well. You will have to choose the information you would like to read. As in the practice session, information about each candidate will appear in a box that scrolls down the screen. If you wish to view a piece of information, click on the box and you will be able to read the contents inside. If you are not interested in some piece of information, you can simply let it scroll down. The electoral campaign will last about 10 minutes.

## Appendix B

### *Secure Scenario*

Breaking news: latest Istat<sup>8</sup> research results

A recent Istat research showed that Italy deeply changed in the last ten years. Crime and delinquency are still present, but decreased so much that every year Italy is becoming more secure than ever. The immigration tension that used to be high at the beginning of the III millennium is over, thanks to some legislative changes approved by a large majority. Today Italy is looked at as an example of harmony and racial integration: Italians and immigrants live and work together contributing to increase social wellbeing. In the world, these are peaceful and flourishing times. For the most part, polls show that Italians feel to live in one of the best periods of the human history, with security, progress, and success spread all over. Tourists are impressed by Italian friendliness, worthiness, integrity, and kindness and by the nice, clean and neat country.

### *Insecure Scenario*

Breaking news: latest Istat research results

A recent Istat research showed that Italy deeply changed in the last ten years. Crime and delinquency are all over and violent assaults take place everywhere. Whenever they can, people avoid walking alone at night because armed squads control many cities districts and go around assaulting and robbing. The huge number of immigrants arrived in Italy in the last years made the situation worse by increasing the crime rates. Home burglaries, especially at night, are today a common experience and are more violent than ever. The police is unable to handle the situation and it seems they are not implementing strategies to fight crime anymore. From the beginning of the III millennium, different governments succeeded, but none of them was able to manage these serious problems.

<sup>8</sup> Istat is the Italian National Institute of Statistics, the main Italian producer of official statistics in the service of citizens and policy makers.